Regression lab: checking residuals

A synthetic teaching handout for RUWithMe

1. Before fitting

Sketch the response against each predictor.

State the assumptions of the linear model.

Keep the simulated data separate from real observations.

2. After fitting

Plot residuals against fitted values.

Inspect leverage and influential observations.

Explain what a curved residual pattern would suggest.

Reading notes and next steps

A synthetic teaching handout for RUWithMe

3. Discussion prompts

Why can a high R-squared coexist with a poor model?

What changes when the error variance depends on a predictor?

Prepare one diagnostic plot for the next recitation.

Space for your observations - no progress is inferred from opening this file.